EE2005 : TUTORIAL - 7 on Synchronous Machines

1. The full load torque angle of a synchronous motor at rated voltage and frequency is 35° (electrical). Neglect the effects of armature resistance and leakage reactance. If the field current is held constant, how would the full-load torque angle be affected by the following changes in operating condition?

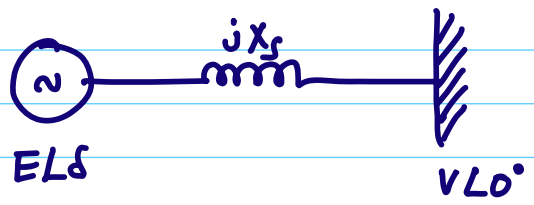
   (a) Frequency reduced by 10 %, load torque and applied voltage constant.

   (b) Frequency reduced by 10 %, load power and applied voltage constant.

   (c) Both frequency and applied voltage reduced by 10%, load torque constant.

   (d) Both frequency and applied voltage reduced by 10%, load power constant.

Sol.



We know

$$|E_\phi| = \frac{|E|}{\sqrt{3}}$$

$$|V_\phi| = \frac{|V|}{\sqrt{3}}$$

Total power output of a Synch. motor.

$$P_o = \left|\frac{EV}{X_s}\right| \sin\delta = 3\left|\frac{E_\phi V_\phi}{X_s}\right| \sin\delta$$

The torque is

$$T = \frac{P_o}{\omega_s}$$

$$T = \left|\frac{EV}{X_s \omega_s}\right| \sin\delta$$

The induced emf E_ϕ

$$E_\phi \propto \omega = 2\pi f_s$$

$f_s \rightarrow$ Supply frequency

$$E_\phi \propto f_s$$

$$\Rightarrow \boxed{E \propto f_s}$$

$$\omega_s = \frac{2\pi N_s}{60}$$

$$= \frac{2\pi}{60} \times \frac{120 f_s}{p}$$

$$\boxed{\omega_s \propto f_s}$$

So

$$\boxed{\begin{aligned} P_0 &\propto \frac{EV}{f_s} \sin\delta \\ T &\propto \frac{EV}{f_s^2} \sin\delta \end{aligned}}$$

(a) $f_2 = (1 - 1 \times 10\%) f_1$

$\boxed{f_2 = 0.9 f_1}$

V = constant

$V_1 = V_2$

$E \propto f$

$$\frac{E_2}{E_1} = \frac{f_2}{f_1} = \frac{0.9 f_1}{f_1} = 0.9$$

$$\boxed{\frac{E_2}{E_1} = 0.9}$$

Given, τ = constant

$$\frac{E_1 V_1}{f_{s_1}^2} \sin\delta_1 = \frac{E_2 V_2}{f_{s2}^2} \sin\delta_2$$

$$\sin\delta_2 = \left(\frac{f_{s2}}{f_{s1}}\right)^2 \left(\frac{E_1}{E_2}\right) \sin\delta_1$$

$$\delta_2 = \sin^{-1}\left(\left(\frac{f_{s2}}{f_{s1}}\right)^2 \left(\frac{E_1}{E_2}\right) \sin\delta_1\right)$$

$$= \sin^{-1}\left(0.9^2 \times \frac{1}{0.9} \times \sin 35^\circ\right)$$

$$\delta_2 = \sin^{-1}(0.516)$$

$$\delta_2 = 31.07895^\circ$$

(b) $f_2 = 0.9 f_1$

V = constant

$\frac{E_2}{E_1} = 0.9$ $\quad P_0$ = constant

$$\frac{E_1 V_1}{f_{s1}} \sin\delta_1 = \frac{E_2 V_2}{f_{s2}} \sin\delta_2$$

$$\sin\delta_2 = \frac{E_1}{E_2} \times \frac{f_{s2}}{f_{s1}} \times \sin\delta_1$$

$$\delta_2 = \sin^{-1}\left(\frac{E_1}{E_2} \times \frac{f_{s2}}{f_{s1}} \times \sin\delta_1\right)$$

$$= \sin^{-1}\left(\frac{1}{0.9} \times 0.9 \times \sin 35^\circ\right)$$

$$= \sin^{-1}\left(\sin 35^\circ\right)$$

$$\boxed{\delta_2 = 35^\circ}$$

δ is unchanged

(c) $f_2 = 0.9 f_1$

applied voltage reduced by 10%

$V_2 = (1 - 1 \times 10\%) V_1 = 0.9 V_1$

$$\frac{E_2}{E_1} = 0.9$$

$$\frac{V_2}{V_1} = 0.9$$

P

Given T = constant

$$\frac{E_1 V_1}{f_{s1}^2} \sin\delta_1 = \frac{E_2 V_2}{f_{s2}^2} \sin\delta_2$$

$$\delta_2 = \sin^{-1}\left(\frac{E_1}{E_2} \times \frac{V_1}{V_2} \times \left(\frac{f_{s2}}{f_{s1}}\right)^2 \times \sin\delta_1\right)$$

$$= \sin^{-1}\left(\frac{1}{0.9} \times \frac{1}{0.9} \times (0.9)^2 \times \sin 35^\circ\right)$$

$$\delta_2 = \sin^{-1}(\sin 35^\circ)$$

$$\delta_2 = 35^\circ$$

δ = unchanged

(d)

$$f_2 = 0.9 f_1$$

$$V_2 = 0.9 V_1$$

$$E_2 = 0.9 E_1$$

Given P_0 = constant

$$\frac{E_1 V_1}{f_{s1}} \sin\delta_1 = \frac{E_2 V_2}{f_{s2}} \sin\delta_2$$

$$\sin\delta_2 = \frac{E_1}{E_2} \times \frac{V_1}{V_2} \times \frac{f_{s2}}{f_{s1}} \sin\delta_1$$

$$\delta_2 = \sin^{-1}\left(\frac{1}{0.9} \times \frac{1}{0.9} \times \frac{0.9}{1} \times \sin 35\right)$$

$$= \sin^{-1}\left(\frac{\sin 35^\circ}{0.9}\right)$$

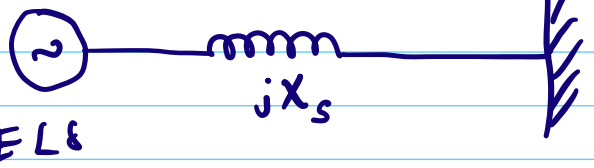
$$\boxed{\delta_2 = 39.591^\circ}$$

2. A three phase star connected alternator is synchronised with an infinite bus of 11 kV. The prime mover input is then increased such that the output power reaches 15 MW. Now, the field excitation is increased to 130%, when the machine starts operating at 0.8 pf lag. Neglecting the armature resistance, determine:

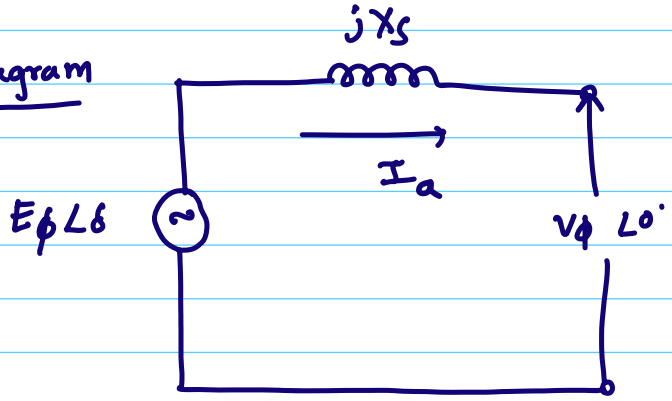
   (a) synchronous reactance of the machine.

   (b) power factor, load angle and armature current before the field excitation was increased.

Sol



per phase diagram



$E_\phi \angle\delta = \frac{E}{\sqrt{3}}\angle\delta$

$V_\phi \angle 0 = \frac{V}{\sqrt{3}}\angle 0$

$$P_o = \frac{|E||V|}{X_s}\sin\delta \quad (\text{or}) \quad = 3\,\frac{|E_\phi||V_\phi|}{X_s}\sin\delta$$

$$P_o = \sqrt{3}\,V_L I_L \cos\phi = 15\,\text{MW}$$

$V_L = V$

$$I_L = \frac{15\times10^6}{\sqrt{3}\times 11\times10^3\times 0.8}$$

$$I_L = 984.119\ \text{A} \quad \text{—(i)}$$

We know

$$E_\phi = \sqrt{(V_\phi \cos\phi + I_a R_a)^2 + (V_\phi \sin\phi + I_a X_s)^2}$$

Let us solve for $I_a X_s$

$$E_\phi^2 = V_\phi^2 \cos^2\phi + V^2 \sin^2\phi + I_a^2 R_a^2 + I_a^2 X_s^2 + 2 V_\phi I_a R_a \cos\phi + 2 V_\phi I_a X_s \sin\phi$$

$(\because R_a = 0)$

$$E_\phi^2 = V_\phi^2 \cos^2\phi + V^2 \sin^2\phi + I_a^2 X_s^2 + 2 V_\phi I_a X_s \sin\phi$$

$$E_\phi^2 = V_\phi^2 + I_a^2 X_s^2 + 2 V_\phi (I_a X_s) \sin\phi$$

$$(I_a X_s)^2 + 2 V_\phi \sin\phi (I_a X_s) + (V_\phi^2 - E_\phi^2) = 0$$

Two roots are

$$I_a X_s = \frac{-2V_\phi \sin\phi \pm \sqrt{2^2 V_\phi^2 \sin^2\phi - 4(V_\phi^2 - E_\phi^2)}}{2 \times 1}$$

$$I_a X_s = -V_\phi \sin\phi \pm \sqrt{V_\phi^2 \sin^2\phi - (V_\phi^2 - E_\phi^2)}$$

Given, $E_\phi = 1.3 \times V_\phi$

$$I_a X_s = -V_\phi \sin\phi \pm \sqrt{V_\phi^2 \sin^2\phi - (V_\phi^2 - 1.3^2 V_\phi^2)}$$

$$I_a X_s = -V_\phi \sin\phi \pm V_\phi \sqrt{\sin^2\phi - (1 - 1.69)}$$

$$I_a X_s = -V_\phi \left[\sin\phi \pm \sqrt{\sin^2\phi + 0.69} \right]$$

given $V_\phi = \frac{11 \times 10^3}{\sqrt{3}}$

$\cos\phi = 0.8 \Rightarrow \sin\phi = 0.6$

then

$$I_a X_s = -\frac{11 \times 10^3}{\sqrt{3}} \times \left(0.6 \pm \sqrt{(0.6)^2 + 0.69} \right)$$

$I_a X_s$ is always +ve

so, $$I_a X_s = -\frac{11 \times 10^3}{\sqrt{3}} \left(0.6 - \sqrt{0.6^2 + 0.69}\right)$$

$$I_a X_s = -\frac{11 \times 10^3}{\sqrt{3}} \times (0.6 - 1.024695)$$

$$= \frac{-11 \times 10^3}{\sqrt{3}} \times (-0.424695)$$

$$I_a X_s = 2697.175985 \text{ A}$$

we already know

$$I_a = 984.119 \text{ A}$$

so $$X_s = \frac{I_a X_s}{I_a}$$

$$X_S = \frac{2697.175985}{984.119}$$

$$X_S = 2.7406988\,\Omega$$

(b)

$$P_0 = 3\,\frac{E_\phi V_\phi}{X_S}\sin\delta \qquad E = V \text{ before excitation was increased}$$

$$= \frac{EV}{X_S}\sin\delta$$

$$\delta = \sin^{-1}\left(\frac{P_0 \times X_S}{E\ V}\right)$$

$$= \sin^{-1}\left(\frac{15\times10^6 \times 2.7406988}{11\times10^3 \times 11\times10^3}\right)$$

$$\delta = 19.862^\circ$$

$$I_a = \frac{E_\phi - V_\phi}{jX_S}$$

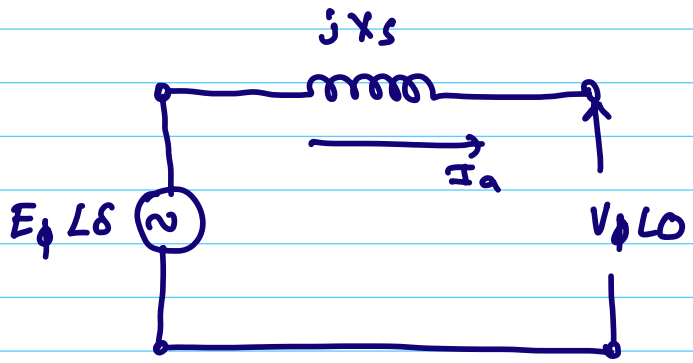
$$= \frac{|E_\phi|\angle\delta - |V_\phi|\angle 0}{jX_S}$$

$$= \frac{|V_\phi|}{X_S}\quad\frac{1\angle\delta - 1\angle 0}{1\angle 90}$$

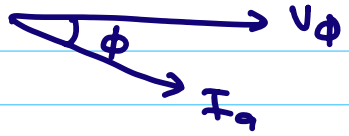
$$= \frac{11\times10^3}{\sqrt{3}\times2.74009} \times \frac{1\angle 19.862 - 1}{1\angle 90}$$

$$I_a = 799.27 \angle 9.931$$

per phase diagram



$|I_a| = 799.274$ A



$pf = \cos\phi = \cos(9.931)$

$= 0.985$ lead

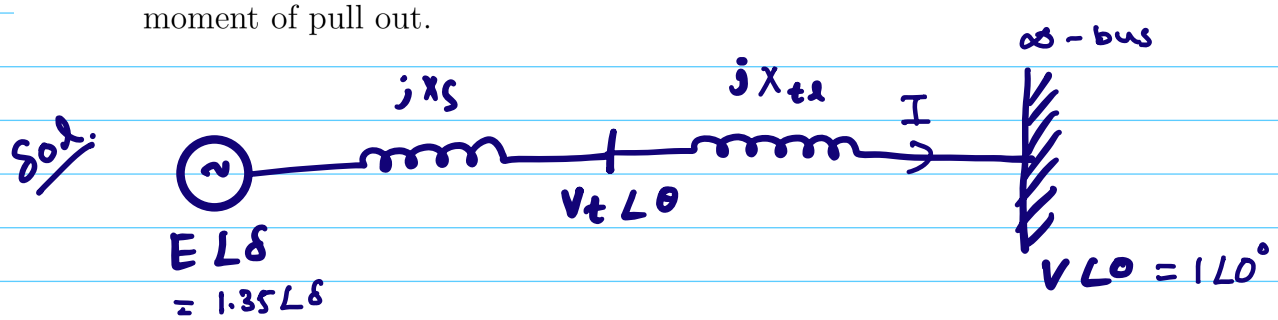
3. A three phase synchronous machine with $X_s = 0.9$ pu on its own rating, is connected via a transmission line to an infinite bus whose voltage is 1 pu. The line reactance is 0.15 pu on the machine base. The internal emf E_f of the machine is kept at 1.35 pu. Calculate

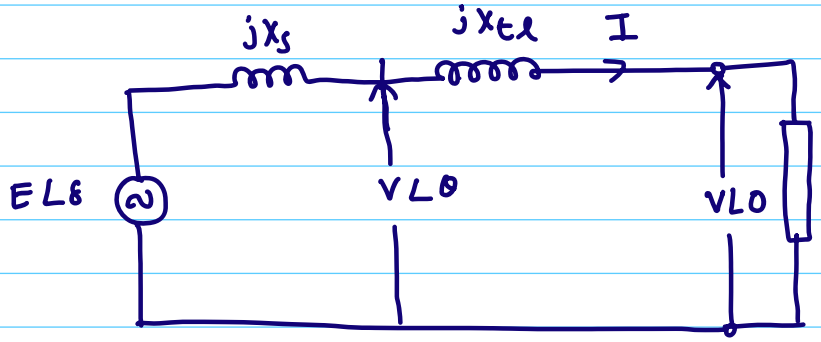
   (a) pull out power of the machine when operating as generator.

   (b) generator terminal voltage and current at pull out.

   (c) reactive powers at the terminals of the generator and at the infinite bus at the moment of pull out.

Sol:





$$P_o = \frac{E\ V}{X_{total}} \sin\delta \quad \left.\right\} \text{ all in p.u.}$$

(a)

pull out power = Max out put power at $\delta = 90°$

$$P_o = \frac{E\ V}{X_{total}}$$

$$= \frac{E\ V}{X_s + X_{tl}} = \frac{1.35 \times 1}{0.9 + 0.15}$$

$$P_o = 1.2857 \text{ pu}$$

(b)

$$I = \frac{E\angle 90 - V\angle 0}{j1.05}$$

$$= \frac{1.35\angle 90 - 1}{j1.05}$$

$$I = 1.6\angle 36.528 \text{ p.u.}$$

Generator terminal voltage V_t,

$$V_t = V + I \times j0.15$$

$$= 1\angle 0 + 1.6\angle 36.528 \times j0.15$$

$$V_t = 0.8786\angle 12.68 \text{ p.u.}$$

(c) Complex power at Gen. terminal

$$S_{gen} = V_t I^*$$

$$= 0.8786\angle 12.68 * 1.6\angle -36.528$$

$$= 1.4057\angle -23.848$$

$$= 1.2857 - j0.5683$$

$$Q_{gen} = -0.5683 \text{ p.u.}$$

Complex power at ∞ bus

$$S_{\infty\text{-bus}} = V I^*$$

$$= 1\angle 0 \times 1.6\angle -36.528$$

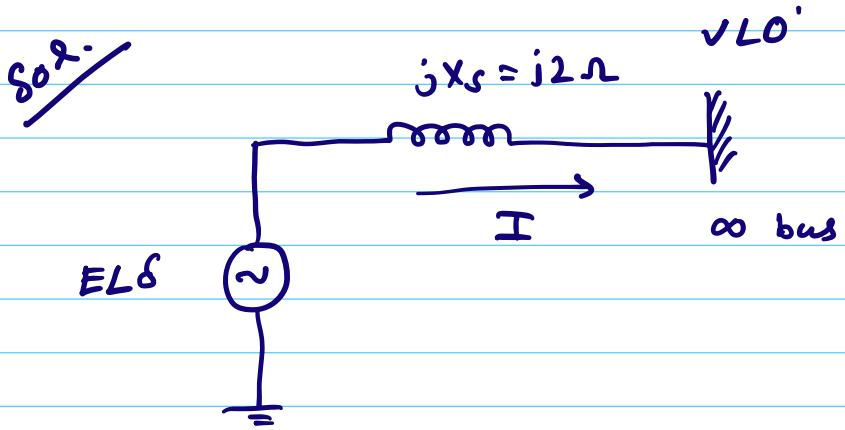
$$= 1.6\angle -36.528$$

$$= 1.2857 - j0.9523$$

$$Q_{\infty\text{-bus}} = -0.9523 \text{ p.u.}$$

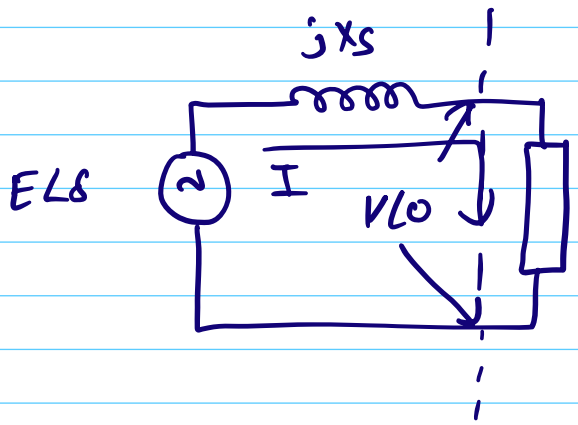
4. A Y-connected synchronous generator connected to an infinite bus at 12 kV supplies 4000 A at unity power factor. The synchronous reactance is 2 Ω per phase and the stator resistance is negligible. If the prime mover set point is unchanged and the field current is increased by 15 %, determine the new stator current and power factor.

Sol^n:



$V = 12\,kV \qquad \bar{V} = 12\times1000\angle 0^\circ\ V$

$V_\phi = \frac{12}{\sqrt{3}} \times 1000\angle 0^\circ\ V$



Given, $I = 4000\angle 0^\circ\ A$

$$E_\phi \angle\delta = \bar{V}_\phi + \bar{I} Z_s$$

$$= \frac{12\times10^3}{\sqrt{3}} + 4000\angle 0^\circ \times j2$$

$$E_{1\phi} \angle\delta = 10583.00524 \angle 49.1066^\circ\ V \text{ (phase)}$$

$|E| \uparrow$ ⇐ Field current, I = 15% ↑

$|E_{2\phi}| = |E_{1\phi}|(1 + 15\%)$

$= 1.15\,|E_{1\phi}|$

$E_{2\phi} = |E_{2\phi}| \angle \delta_2$

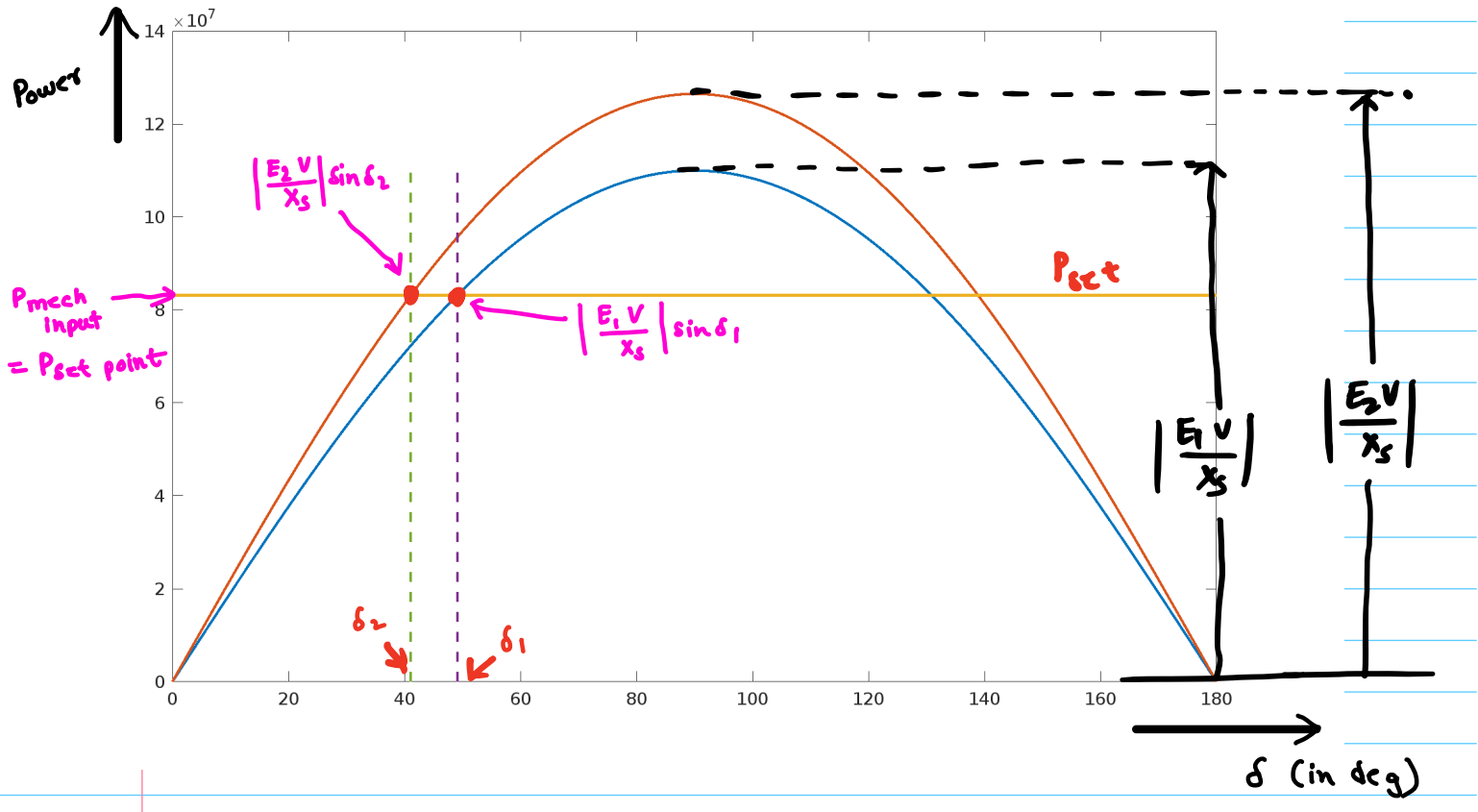
$= 1.15\,|E_{1\phi}| \angle \delta_2$

$= 1.15\,E_{1\phi} \angle \delta_2$

$E_{2\phi} = 12170.45603 \angle \delta_2$ (phasor)

Given Prime Mover set point is Unchanged, so

$$P_{mech.input} = \left|\frac{E_1 V}{X_s}\right| \sin\delta_1 = \left|\frac{E_2 V}{X_s}\right| \sin\delta_2$$



$$\frac{E_1 V}{X_S}\sin\delta_1 = \frac{E_2 V}{X_S}\sin\delta_2$$

E_1, E_2, V } Line quantities

given

$|E_2| = 1.15|E_1|$

or

$|E_{2\phi}| = 1.15|E_\phi|$

$$\delta_2 = \sin^{-1}\left(\frac{E_1}{E_2}\sin\delta_1\right)$$

$$= \sin^{-1}\left(\frac{E}{1.15E_1}\sin\delta_1\right)$$

$$= \sin^{-1}\left(\frac{\sin(49.1066^\circ)}{1.15}\right)$$

$$\delta_2 = 41.09652^\circ$$

$$E_{2\phi} = 12170.45603 \angle 41.09652 \text{ V} \quad \text{(phase)}$$

$$I_2 = \frac{E_{2\phi} - V_\phi}{jX_S}$$

$$= \frac{12170.4563\angle 41.09652 - \frac{12000}{\sqrt{3}}\angle 0}{j2}$$

$$I_2 = 4154.31275 \angle -15.6654$$

$$|I_2| = 4154.3127 \text{ A}$$

$$pf = \cos\phi = \cos(15.6654) = 0.96285 \text{ lagging}$$

5. A 4-pole, 60 Hz, 24 kV, 650 MVA synchronous generator with a synchronous reactance of 1.82 per unit is connected to a grid which can be represented by a 24 kV infinite bus in series with a reactive impedance of j 0.21 Ω. The generator is equipped with a voltage regulator that adjusts the field excitation such that the generator terminal voltage remains at 24 kV independent of the generator loading. The output power of the generator is adjusted to 375 MW.

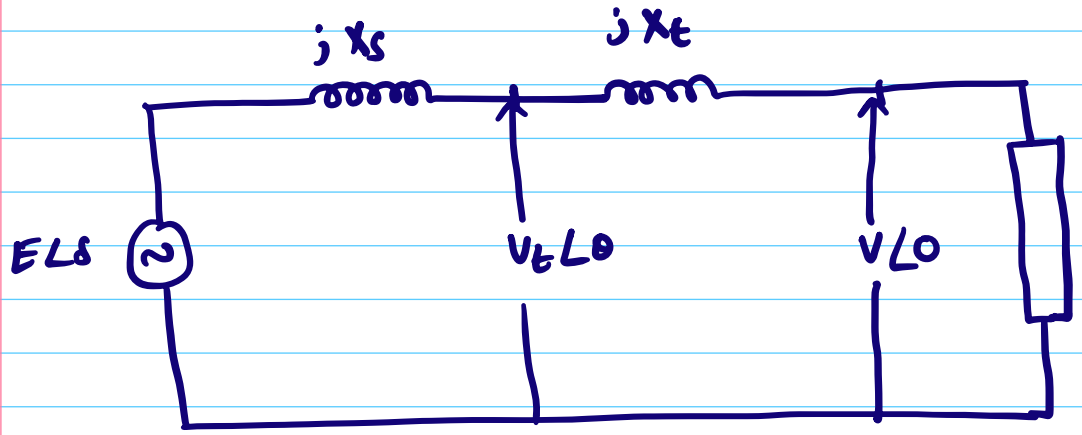
   (a) Draw a phasor diagram for this operating condition.

   (b) Find the magnitude and phase angle (with respect to the generator terminal voltage) of the generator terminal current.

   (c) Determine the generator terminal power factor.

   (d) Find the magnitude (in p.u and kV) of the generator excitation voltage E_f .

Sol.



Base values

$V_{Base} = 24kV$

$S_{Base} = 650MVA$

$S = \sqrt{3}\, V_L I_L$

$I_L = \frac{S}{\sqrt{3}\, V_L}$

$I_L = I_{Base} = \frac{650M}{\sqrt{3} \times 24 \times 10^3}$

$I_{Base} = 15636.56979\ A$

$$Z_{Base} = \frac{V_{Base}^2}{S_{Base}} = \frac{V_{Base}^2}{V_{Base} * I_{Base}}$$

$$= \frac{(24 \times 10^3)^2}{650 \times 10^6}$$

$$= 0.88615\ \Omega$$

$$X_t\ pu = \frac{0.21}{0.8862}$$

$$X_t = 0.23697\ pu$$

In p.u. values

$|V_t| = 24kV$, $|V| = 24kV$ $\longrightarrow$ $|V_t| = 1$ p.u., $|V| = 1$ p.u.

$$P_o\ (p.u.) = \frac{P_o}{S_{base}}$$

$$= \frac{375 \times 10^6}{650 \times 10^6}$$

$$= 0.5769\ p.u.$$

$$P_o = \frac{V\ V_t}{X_s} \sin\delta \quad \text{(all in p.u. values)}$$

$$\delta = \sin^{-1}\left(\frac{P_o \times X_s}{V\ V_t}\right)$$

$$= \sin^{-1}\left(\frac{0.5769 \times 0.23697}{1 \times 1}\right)$$

$$\delta = 7.8577^\circ$$

So, the generator terminal voltage, V_t

$$V_t = 1 \angle 7.8577^\circ\ p.u.$$

with actual values

$$\frac{V_t\ V}{X_t} \sin\theta = 375MW$$

$$\theta = \sin^{-1}\left(\frac{375 \times 10^6 \times 0.21}{(24 \times 10^3)^2}\right)$$

$$\theta = 7.858^\circ$$

(b)

$$I = \frac{V_t - V}{jX_t}$$

$$= \frac{1\angle 7.858 - 1\angle 0}{j\,0.2369}$$

$$= 0.57828\angle 3.929^\circ \text{ p.u.}$$

$$I = 0.57828\angle 3.929^\circ \text{ p.u.}$$

$$|I| = 0.57828 \text{ p.u.}$$

(c) power factor between $|V_t|\angle\theta$ & $|I|\angle\phi$

$$pf = \cos(\theta - \phi) = \cos(7.858 - 3.929)$$

$$= 0.9976 \text{ lag}$$

$$pf = 0.9976 \text{ lag}$$

(d) Generator internal generated emf, E_f

$$E_f = V_t + I * jX_s$$

$$= 1\angle 7.858 + 0.5783\angle 3.929 \times j1.82$$

$$E_f = 1.5\angle 52.26^\circ \text{ p.u.}$$

E_f in k.V.

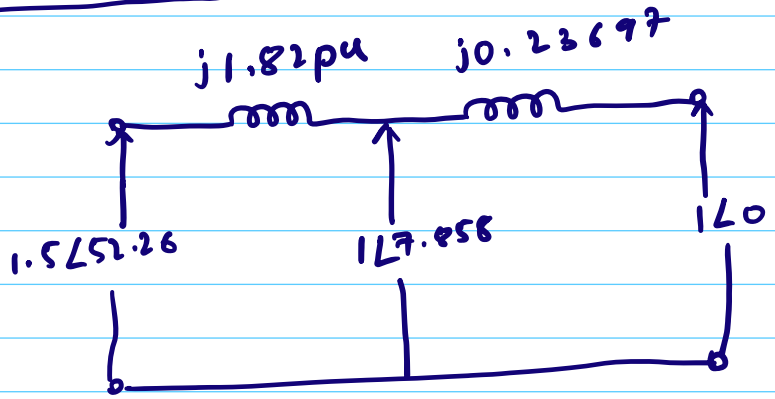
$$E_f = E_f \text{ (p.u.)} \times V_{Base}$$

$$= 1.5\angle 52.26 \times 24 \times 10^3$$

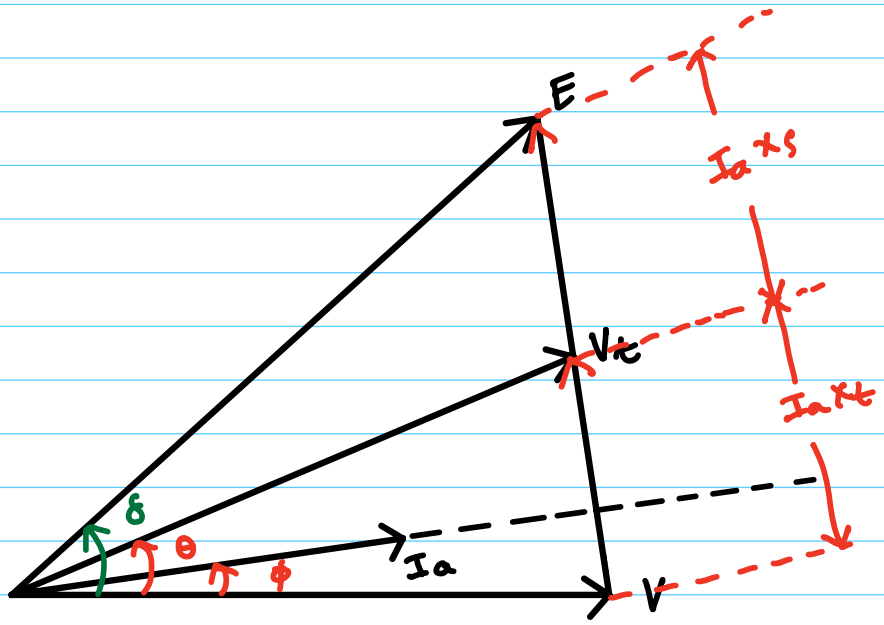
$$E_f = 36\angle 52.26 \text{ KV} \quad \text{(Line voltage)}$$

$$|E_f| = 36 \text{ KV} \quad \text{(Line voltage)}$$

per phase diagram:



(a)



Phasor Diagram with p.u. values